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Empirical Finance



Does betting on Standardized Unexpected Earnings work? Revisiting the Post-Earnings Announcement Drift

An Analysis by

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Executive Summary

This report examines whether betting on Standardized Unexpected Earnings (SUE) still yields abnormal returns in today's financial markets, revisiting the Post-Earnings Announcement Drift (PEAD) anomaly described by Bernard and Thomas (1989). Using data from 2004 to 2023, we analyse earnings announcements from U.S. companies listed on NASDAQ, NYSE, and AMEX. We construct long-short portfolios by buying stocks in the top SUE decile and shorting those in the bottom SUE decile, measured by the realized Cumulative Abnormal Returns (CAR), over a period of 60 trading days following each earnings announcement.

Our results show that, on average, the strategy does not produce statistically significant abnormal returns across the full sample. This suggests that PEAD has weakened over the last two decades. However, further analysis reveals that PEAD still exists in large-cap stocks, contradicting findings from prior research. At the sector level, a few industries generate positive abnormal returns, while other industries show a significant negative performance.

Therefore, in this study we argue that while PEAD is no longer broadly exploitable, SUE-based strategies remain relevant when applied selectively based on firm size and sector characteristics.

Table of Contents

1. Introduction	1
2. Relevance and Motivation of the Study.....	2
3. Data and Methodology	3
4. Results.....	6
5. Conclusion	12
Disclaimer.....	13
Bibliography.....	13

1. Introduction

Our report revolves around the phenomenon of the post-earnings-announcement drift (PEAD), as observed by Bernard and Thomas (1989)¹. The post earnings announcement drift represents a persistent anomaly in financial markets, first noted by Ball and Brown (1968)² and confirmed by Foster et al. (1977)³. This anomaly suggests that publicly available earnings information is not immediately incorporated into stock prices. In their paper "Post-Earnings-Announcement Drift: Delayed Price Response or Risk Premium?", Bernard and Thomas (1989) find that stock prices continue to drift in the direction of unexpected earnings long after earnings announcements have been made. Bernard and Thomas examine whether this anomaly can be attributed to a delayed market reaction to public information or instead reflects a mismeasurement of risk premiums within asset pricing models.

Their study provides robust empirical evidence that the drift cannot be fully explained by traditional risk-based models, including adjustments to beta or exposure to macroeconomic risk factors. Instead, the results are more consistent with a gradual assimilation of earnings information into market prices, possibly due to investor underreaction or implementation frictions. This insight directly challenges the efficient market hypothesis in its semi-strong form and has since become a cornerstone reference in empirical asset pricing and behavioral finance. Later works, such as those by Freeman and Tse (1989)⁴ and Ou and Penman (1989)⁵, further reinforce the idea that markets do not fully anticipate the implications of earnings information for future performance.

The continued relevance of this research lies in its implications for both academic finance and investment practice. For scholars, the PEAD anomaly invites further refinement of asset pricing models, especially in incorporating behavioral elements. For practitioners, it offers empirical grounding for earnings-based investment strategies that have demonstrated persistent abnormal returns.

Our research adds to Bernard and Thomas's findings, through expanding the observed timeframe to recent history, using Stata, examining if the post earnings announcement drift persists in times of market accessibility. Moreover, we expand by evaluating if the observed post earnings announcement drift correlates with the company's size and industry.

This report is structured as follows. Section 2 discusses the financial problem and the underlying theoretical models. Section 3 outlines the data and methodology employed to replicate and extend the original findings. Section 4 presents the empirical results, and Section 5 concludes with implications and suggestions for future research.

¹ (Bernard & Thomas, 1989)

² (Ball & Brown, 1968)

³ (Foster, 1977)

⁴ (Freeman & Tse, 1989)

⁵ (Ou & Penman, 1989)

2. Relevance and Motivation of the Study

The Post-Earnings Announcement Drift (PEAD) is still one of the most well-known and studied anomalies in finance. It refers to the tendency of stock prices to keep moving in the direction of an earnings surprise even after that information has been made public. Early research, like the work by Bernard and Thomas in 1989, found strong evidence supporting this effect. But an open question remains: does this anomaly still exist today, in a world where markets are faster, more efficient, and more technologically advanced?

Since the 1980s, financial markets have changed drastically. Algorithmic trading has taken off, financial news spreads instantly, and investors have become more informed and sophisticated. In this new environment, a slow market reaction to public earnings news, like we see with PEAD, challenges the idea that markets efficiently reflect all available information, as suggested by the semi-strong form of the Efficient Market Hypothesis (EMH).

In our study, we take a fresh look at the PEAD anomaly using modern data from the period 2004 to 2023. Unlike most past studies, which mainly tried to explain the anomaly through theories about investor behavior or risk, our approach is straightforward, and data driven. We want to find out whether PEAD still exists in today's markets, and, if it does, under what conditions it shows up more clearly.

We focus in particular on how the size of a company (measured by market capitalization) and its industry sector might affect the size and duration of post-earnings price movements. These factors could matter a lot: for example, smaller companies often receive less attention from analysts and investors, which might delay how quickly their earnings information is priced in. At the same time, different sectors, depending on their complexity, regulations, or business models, might lead investors to react differently to earnings surprises.

By grouping companies based on their size, we investigate whether PEAD is still a widespread pattern in the market or whether it's now more limited to certain kinds of firms. Our idea is that smaller companies might still show slower reactions, while larger, more closely followed firms might adjust more quickly. Similarly, sector-related differences could affect how earnings surprises are interpreted and absorbed by investors.

To test all of this, we use a classic event-study approach. Following the original design by Bernard and Thomas, we build a trading strategy based on what's called Standardized Unexpected Earnings (SUE) and look at how stock prices behave after earnings announcements across different types of companies. We explain our data, the way we calculate SUE, and how we build portfolios in the next section.

Finally, we measure the market response using Cumulative Abnormal Returns (CAR) over a 60-day window after each earnings announcement. While the original studies looked at much longer periods (up to 240 days), we focus on the first 60 trading days, which is when most of the price movement tends to happen. This also matches what more recent research and real-world investors usually focus on.

This study seeks to answer the following questions: (i) Does PEAD still exist in modern financial markets? (ii) Does the drift vary by firm size? (iii) Are certain industry sectors more prone to this anomaly?

3. Data and Methodology

This study analyzes the post-earnings-announcement drift using firm-level data from CRSP and IBES over a 20-year period. The dataset includes quarterly earnings, stock prices, SIC industry codes, and market capitalization data for U.S. publicly traded firms.

The methodology consists of four core components: (i) construction of standardized unexpected earnings (SUE) as the earnings surprise measure, (ii) decile portfolio formation based on lagged SUE, (iii) measurement of cumulative abnormal returns (CARs), and (iv) implementation of a long-short investment strategy.

We retrieve index constituents from the New York Stock Exchange (NYSE), American Stock Exchange (AMEX), and National Association of Securities Dealers Automated Quotations (NASDAQ). We collect all listed firms from these exchanges between January 2004 and December 2023, resulting in approximately 14,000 unique tickers.

These tickers are matched with the IBES database to extract quarterly EPS values, announcement dates, fiscal period ends, and quarter identifiers over the 2004–2023 period, resulting in approximately 248,000 observations.

The collected data was used to compute Standardized Unexpected Earnings (SUE). First, the dataset is cleaned by removing duplicate observations based on the unique combination of tickers and announcement dates. The period end date and announcement dates are converted to Stata internal date format to facilitate time-series operations. Each ticker is then encoded into a unique numeric identifier to enable grouped operations. The data is sorted by these identifiers and announcement date to establish a chronological order. A sequential time index is created within each ticker group to track the position of each observation in the time series. Lagged variables are generated: A variable for 1-period (L1.value), 4-periods (L4.value), and 5-periods (L5.value) lags of the actual EPS values. These lags are used to compute two variation metrics: The first measures the difference between the current and the 4-quarter lagged value (variation_1_year), and the second measures the difference between the 1-quarter lagged and 5-quarter lagged values (variation_1lag).

In the following, we run a 24-period rolling regression of the one-year difference on the one-year difference lagged by a quarter for each ticker. Regression outputs are cleaned by removing all variables except the coefficients. The coefficient on difference between the 1-quarter lagged and 5-quarter lagged values (variation_1lag) is renamed beta, and the intercept is renamed alpha. Using these coefficients, the model computes a forecasted value as:

$$\widehat{EPS}_t = EPS_{t-4} + \hat{\alpha} + \hat{\beta} \cdot (EPS_{t-4} - EPS_{t-5}) + \varepsilon_t$$

For observations with fewer than 16 prior data points, the model replaces the forecast with the 4-lag value ($\widehat{EPS}_t = EPS_{t-4}$), thus assuming the predicted value follows a random walk. This is also done to maintain consistency and mitigate estimation bias. After computing the

forecasted EPS, we define the forecast error as the difference between the realized earnings and the model-implied forecast. The standard deviation of these forecast errors is estimated using a 24-quarter rolling window to ensure sufficient historical variability.

$$\sigma(\hat{\varepsilon}) = EPS_t - \widehat{EPS}_t$$

The standardized unexpected earnings (SUE) is then calculated as:

$$SUE_t = \frac{EPS_t - \widehat{EPS}_t}{\sigma(\hat{\varepsilon})}$$

To mitigate the influence of extreme observations, the SUE variable is winsorized at the 1st and 99th percentiles. This process sets any values below the 1st percentile to the 1st percentile value, and any values above the 99th percentile to the 99th percentile value, reducing the influence of outliers on subsequent analysis.

We retrieve monthly data on share price and number of shares outstanding to calculate market capitalization for firm size and SIC industry codes for all firms over the same 20-year period. The dataset comprises 1,048,576 observations.

Based on size, the companies are ranked in deciles, to distinguish small-cap, medium-cap and large-cap stocks. This allows for dynamic, cross-sectional comparisons of firm performance by size. The SIC codes classification enables sector-level control in empirical analysis and supports the construction of industry-adjusted return portfolios.

For the calculation of the Cumulative Abnormal Returns (CAR), an event study was conducted on CRSP, based on a list of tickers and announcement dates. The abnormal returns are calculated using the market-adjusted model, with the following conditions: estimation window of 60 days, 40 minimum number of valid returns, 60 days gap between estimation window and event study, start of event window of 1 day before the event and 60 days of event window after the event. The file was subsequently cleaned, dropping double values and multiple observations for the same date and ticker. The event study yields approximately 388,000 cumulative abnormal return (CAR) observations. Moreover, the event study provided us the average evolution of the CAR in the time window in Figure 1.

Cumulative Abnormal Return: Mean & 95% Confidence Limits

There are 388599 events in total with non-missing returns.

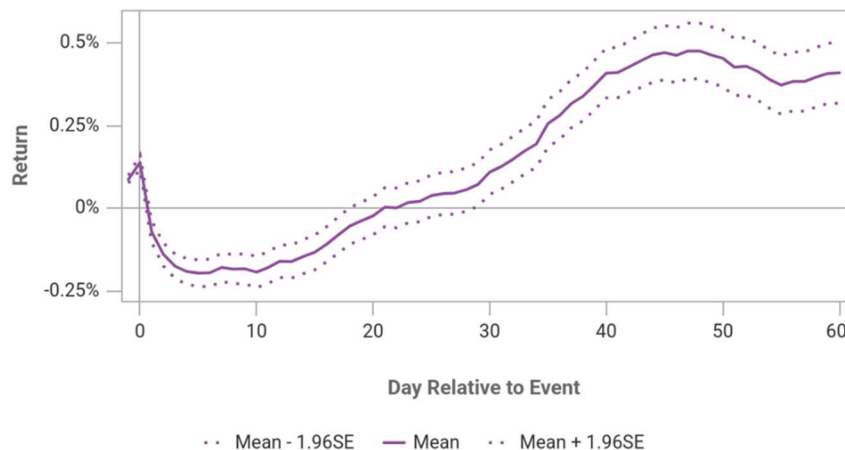


Figure 1 – average CAR trajectory during event window.

The CAR data is merged with size and industry by matching ticker and date. The new merged file resulted in 220,000 observations. The new file is subsequently merged with the SUE file, based on the exact match of announcement date and ticker variables. After the merged file is cleaned, it has 160,000 observations.

To classify firms based on the cross-sectional distribution of their standardized unexpected earnings (SUE), each observation is assigned to a decile within its corresponding calendar quarter. This procedure ensures that decile ranks are computed relative to firms reporting in the same period, preserving temporal comparability. SUE values are computed immediately upon the earnings announcement. Each stock is assigned to a SUE decile immediately upon earnings announcement, using decile cutoffs determined from the cross-sectional distribution of SUE in the previous quarter, ensuring that portfolio formation is forward-looking and implementable. This ensures that portfolio formation reflects an implementable, real-time strategy. The cross-sectional ranking is performed as follows: firms in the highest 10% of the SUE distribution (top decile) are assigned to Decile 10 (D10); firms in the lowest 10% of the SUE distribution (bottom decile) are assigned to Decile 1 (D1); the remaining firms are assigned to intermediate deciles (D2 through D9) accordingly. Each firm can appear in multiple quarterly decile rankings over the sample period, depending on the timing and magnitude of its earnings surprises.

$$D_{i,t} \in \{1, 2, \dots, 10\}$$

$$D_{i,t} = d \text{ if } SUE_{i,t} \in [Q_{d-1}^{(t)}, Q_d^{(t)})$$

For the purpose of the whole portfolio strategy (all assets), we took into account the companies that, throughout the years, were assigned to the first or the tenth deciles.

Unlike traditional calendar-based portfolio rebalancing, our strategy operates on a rolling, event-driven basis. Each firm enters the portfolio individually on the day of its earnings announcement, contingent on its SUE ranking. Stocks are included in the long portfolio if their SUE places them in Decile 10, and in the short portfolio if they fall into Decile 1, based on cross-sectional cutoffs from the previous quarter. Positions are weighted equally at entry. To approximate equal weighting over time, we estimate the average number of positions per decile per quarter (N) and assign a fixed weight of $1/N$ to each newly added position. Returns are then accumulated for each position over a 60-day post-announcement window. This structure allows for continuous updating of the portfolio and reflects the asynchronous nature of earnings releases across firms.

The overall performance of the portfolio is, eventually, calculated as the difference in the average CARs:

$$\text{Return}_{\text{Long-Short}, t} = \text{CAR}_{\text{D10}, t} - \text{CAR}_{\text{D1}, t}$$

To test statistical relevance, using Stata, we computed the summary statistics (mean, standard deviation, min, max) of the annualized returns. Furthermore, A two-sided t-test is performed to evaluate whether the average annualized return differs significantly from zero. For each long-short portfolio, we also compute the Sharpe ratio as the mean excess return divided by the standard deviation of returns, using daily CARs aggregated across the event window.

To examine whether the post-earnings announcement drift varies across sectors and to control potential industry-specific effects, we replicated the portfolio formation procedure within selected industry groupings, based on the number of observations in our sample. Within each industry, we applied the same methodology described in the previous sections. Furthermore, we compare the magnitude and statistical significance of the long-short decile spread across industries to assess heterogeneity in the strength of the PEAD effect.

To investigate whether the post-earnings announcement drift differs across firms of varying size, we implement a long–short trading strategy within separate market capitalization groups. Firms are classified into small-cap, mid-cap, and large-cap segments based on their monthly market capitalization. Within each group, we construct portfolios by going long on stocks in the highest SUE decile and short on those in the lowest SUE decile. This procedure replicates the methodology applied in the overall sample, ensuring consistency across groups. By comparing the performance and statistical significance of the decile spread across size segments, we assess whether the strength of the PEAD effect is systematically related to firm size.

4. Results

4.1 Standardized Unexpected Earnings

Table 3 displays the standardized unexpected earnings per decile.

Table 3: SUE per Decile						
d	obs.	Mean	SD	Min	Max	Kurt
1	15,837	-2.105	1.780	-6.251	-0.190	3.560
2	15,862	-0.460	0.350	-2.790	-0.300	9.550
3	15,830	-0.170	0.170	-1.102	0.500	10.010
4	15,886	-0.350	0.100	-0.570	0.200	10.330
5	15,776	0.520	0.090	-0.290	0.420	5.590
6	15,855	0.160	0.130	-0.130	0.790	4.270
7	15,832	0.350	0.240	-0.020	1.390	3.640
8	15,855	0.650	0.410	0.030	2.220	3.230
9	15,804	1.200	0.720	0.090	3.800	2.800
10	15,768	3.940	5.510	0.270	12.600	4.270

The lowest decile exhibits a negative average SUE of -2.105 and high dispersion, while the highest decile shows the strongest positive average SUE of 3.940 and the largest standard deviation 5.510 . The distribution is highly leptokurtic in the middle deciles, particularly Deciles 3 and 4, indicating concentrated but extreme deviations around the center. Moreover, we display the average SUE for the top and bottom decile from 2005 to 2025 in Figure 2.

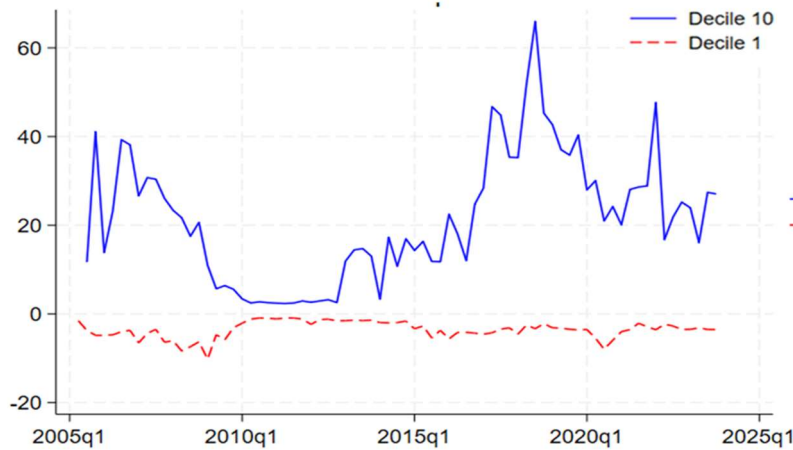


Figure 2 – Evolution of average SUE for Decile 10 and Decile 1

Figure 2 shows high volatility in the average SUE in Decile 10 (positive surprises), with multiple spikes—most notably around 2020—peaking above 60. In contrast, Decile 1 (negative surprises) remains stable and consistently close to zero throughout the entire period. This divergence suggests that the magnitude of strong positive earnings surprises varies significantly over time, while negative surprises are relatively uniform.

4.2 Standardized Unexpected Earnings by Size

Table 4: SUE by size (Decile 10)

Market cap decile	obs.	Mean	Std	MIN	MAX	t-test	p-value
Small-cap	2,757	-0.043	0.457	-0.982	0.993	-4.94	0.00
Medium-cap	3,405	-0.003	0.455	-0.984	0.972	-0.43	0.66
High-cap	4,419	0.015	0.452	-0.970	0.994	2.27	0.02

To capture the firm specific effects on the SUE and their effects, we focus on the SUE of the top and bottom decile, categorized by size. In Table 4, only the high-cap portfolio in Decile 10 yields a positive and statistically significant average SUE, with a t-statistic of 2.27 and a p-value of 0.02. In contrast, small-cap portfolios show significantly negative SUEs.

Table 5: SUE by size (Decile 1)

Market cap decile	obs.	Mean	Std	MIN	MAX	t-test	p-value
Small-cap	3,538	-0.017	0.473	-0.989	0.998	-2.08	0.04
Medium-cap	6,079	0.001	0.463	-0.996	1.000	0.12	0.91
High-cap	1,935	-0.016	0.443	-0.964	0.971	-1.54	0.12

In Table 5, the small-cap portfolio in Decile 1 exhibits a statistically significant SUE, with a p-value of 0.04. Compared to Decile 10, the mean SUEs are less significant in our Decile 1 bracket. High-cap stocks show significant positive SUEs, suggesting the evidence that the informativeness of extreme SUE values varies by size: small firms react more to negative surprises, while large firms react more to positive ones.

4.3 Long-Short Strategy by Size

In the following we implement our prior findings in long Decile10, short Decile 1, according to their size. To illustrate the evolution of CAR by size, we introduce Figure 3.

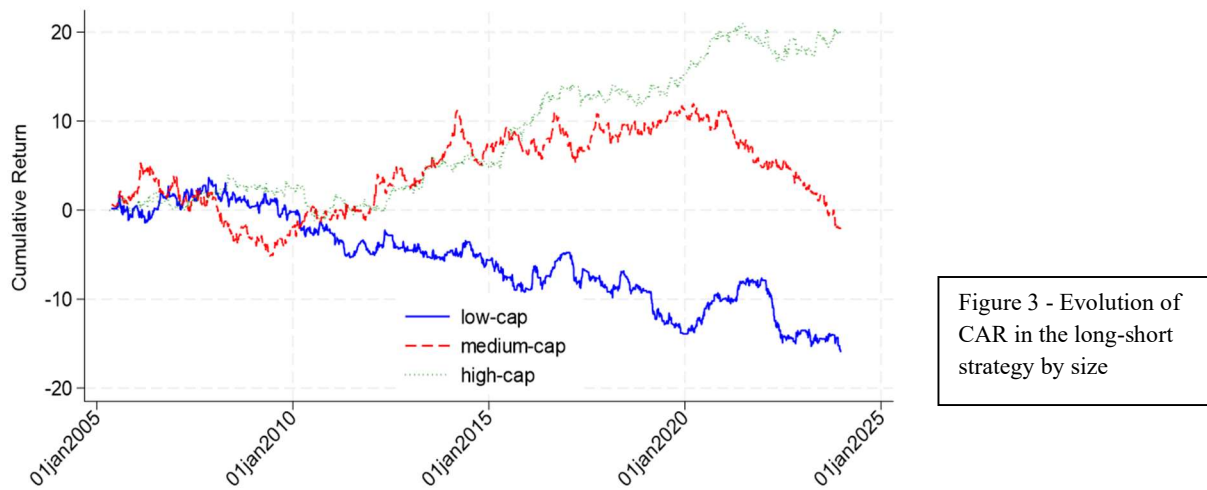


Figure 3 shows the cumulative return of a long-short SUE strategy over time, separated by firm size categories. The strategy underperforms consistently in the low-cap segment, showing a steady decline, reaching approximately -20% by 2023. The medium-cap portfolio initially trends upward but reverses after 2020, showing no positive cumulative return in the observed period. In contrast, the high-cap segment generates strong and persistent positive returns, peaking near 20% before a slight drawdown post-2022. Table 6 provides the corresponding descriptive statistics.

Size	Obs	Mean	Std	Min	Max	SR	t-test	p-value
Small-cap	2,205	-0.72%	22.38%	-116.57%	100.40%	-1.52	-1.52	0.13
Medium-cap	2,608	-0.09%	22.56%	-98.11%	119.93%	-0.20	-0.20	0.84
High-cap	2,047	0.96%	21.50%	-93.20%	106.85%	2.02	2.02	0.04

In Table 6, only the high-cap portfolio yields a statistically significant mean return of 0.96% . In contrast, the small-cap and medium-cap portfolios show negative returns. This contradicts the findings from prior research that found the strategy provides the smallest and most insignificant return for High-cap firms and stronger PEAD effects among smaller firms. This suggests a shift in the size effect over time, with large firms now exhibiting a more exploitable post-earnings drift.

4.4 Standardized Unexpected Earnings by Industry

To capture the firm specific effects on the SUE and their effects, we focus on the SUE of the top and bottom decile, categorized by industry. Table 7 shows the SUEs for the top decile.

Table 7: SUE by industry (Decile 10)

Industry	obs.	Mean	SD	MIN	MAX
Durbl	251	2.85	5.11	0.33	59.50
Enrgy	683	4.15	10.71	0.30	112.90
HiTec	2,120	16.76	110.06	0.30	1,888.00
HLth	1,992	32.91	146.26	0.29	1,832.00
Manuf	1,886	9.47	72.59	0.27	1,530.00
NoDur	772	8.35	67.67	0.30	1,620.00
Other	5,343	38.79	180.24	0.29	1,953.60
Shops	1,720	8.60	70.20	0.31	1,800.00
Telcm	196	14.30	75.00	0.30	770.00
Utils	559	4.94	32.88	0.31	770.50

Table 7 shows that the magnitude of SUEs in the top decile varies substantially across industries. The Other and Health industries exhibit the highest average SUEs, with means of 38.79 and 32.91, respectively, and standard deviations exceeding 140. In contrast, industries like Durbl and Energy display low mean values below 5 and narrow dispersion.

Table 8: SUE by industry (Decile 1)

Industry	obs.	Mean	SD	MIN	MAX
Durbl	451	-3.58	11.44	-155.00	-0.21
Enrgy	1,115	-3.18	8.74	-157.15	-0.21
HiTec	1,780	-3.00	10.37	-180.00	-0.19
HLth	968	-7.78	25.37	-198.00	-0.19
Manuf	1,863	-2.60	8.03	-172.80	-0.21
NoDur	698	-2.67	9.86	-180.00	-0.20
Other	6,529	-4.04	12.50	-192.00	-0.19
Shops	1,376	-3.30	9.36	-156.00	-0.19
Telcm	355	-3.69	12.85	-140.00	-0.20
Utils	379	-2.55	7.85	-100.00	-0.22

Table 8 shows the SUE by industry in the bottom decile and presents similar findings to Table 7. The Health and Other industry exhibit the most extreme mean SUEs (-7.78; -4.04) and the highest standard deviation (25.37; 12.50). In contrast, industries like Manuf, NoDur, and Utils report relatively mild negative surprises, with means above -3 and lower dispersion, indicating that the strength and volatility of negative SUE signals remain concentrated in specific sectors with high informational volatility.

4.5 Long-Short strategy by industry

In the following we implement our prior findings in long Decile10, short Decile 1, for the entire portfolio, and according to their industry. To illustrate the evolution of CAR by size, we introduce Figure 4. Table 9 provides the corresponding descriptive statistics.

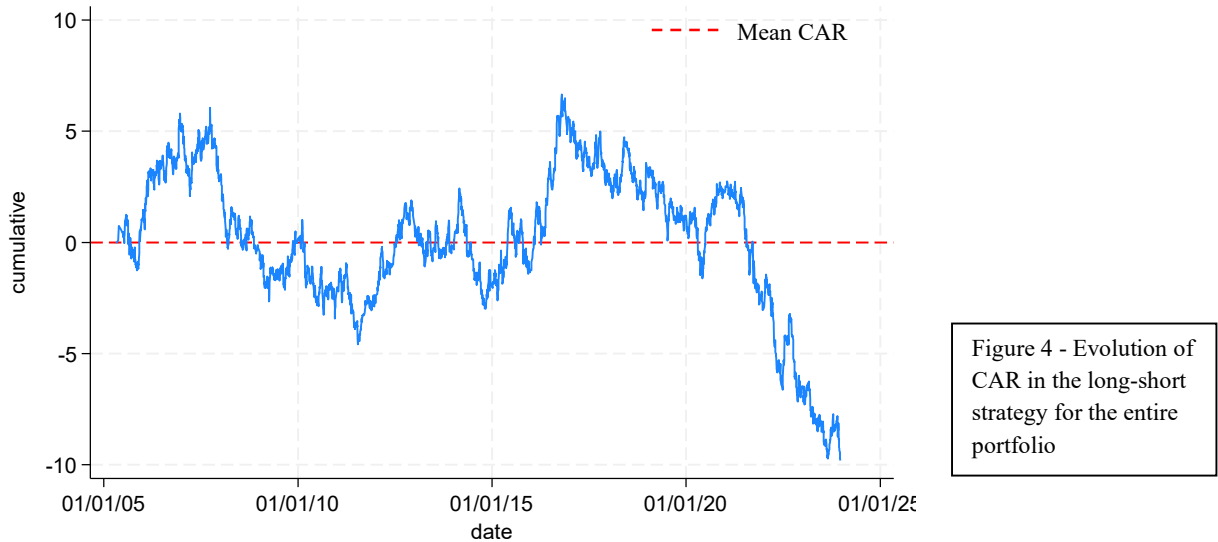


Figure 4 shows the cumulative abnormal return (CAR) of the long-short strategy applied to the full portfolio over time. The strategy fluctuates around zero, showing intermittent gains without clear sustained momentum. However, from 2020 onward, the portfolio enters a persistent decline, reaching nearly -10% by early 2025. This pattern indicates that the strategy has lost effectiveness in recent years, and overall performance across the full sample does not generate positive long-term returns.

Table 9: Long Short by Industry

Industry	Obs	Mean	Std	Min	Max	SR	t-test	p-value
All	13,324	-0.26%	20.08%	-1.1368	1.1304	-1.49	-0.79	0.43
Dur	614	2.27%	45.97%	-0.9140	0.9514	1.22	1.22	0.22
Energy	928	-1.87%	45.79%	-0.8595	1.0590	-1.24	-1.24	0.21
Health	1,457	0.15%	44.34%	-1.1073	0.9783	0.13	0.13	0.90
HiTech	1,838	2.82%	50.62%	-1.1368	1.1057	2.39	2.39	0.02
Manuf	1,810	2.71%	47.90%	-0.9961	1.1304	2.41	2.41	0.02
NoDur	1,125	2.26%	47.48%	-0.9599	1.0109	1.60	1.60	0.11
Other	2,679	-2.17%	43.70%	-1.0818	0.9942	-2.57	-2.57	0.01
Shops	1,870	2.89%	46.13%	-0.9312	1.0134	2.71	2.71	0.01
Telcom	445	-0.76%	45.24%	-0.8856	0.7094	-0.36	-0.36	0.72
Utils	558	-4.33%	45.13%	-0.8850	0.9367	-2.26	-2.26	0.02

Table 9 shows that the long-short strategy yields statistically significant positive returns of 2.8%, 2.7% in HiTech, Manuf, and Shops, respectively. In contrast, the strategy yields significant negative returns in the Other and Utils sectors, both with p-values of 0.02, and in each case accompanied by Sharpe ratios below -2 . The overall portfolio does not produce significant results, with a mean return of -0.26% and a p-value of 0.43. These findings indicate that the profitability of SUE-based long-short strategies is driven by specific sectors, while a broad-market implementation fails to generate consistent excess returns.

4.6 Discussion of Results

The results confirm the remaining relevance of the post-earnings announcement drift (PEAD), but challenge the broad application of a standardized unexpected earnings (SUE) based trading strategy. Bernard and Thomas (1989) document that PEAD is the strongest and most statistically significant among small-cap stocks, a finding supported by their evidence of delayed price adjustment and investor underreaction. In contrast, our results show that this size effect no longer holds and seems to be opposite: the long-short strategy is only effective in the high-cap segment, where we observe statistically significant positive returns. Small- and mid-cap firms do not exhibit exploitable drift under the same methodology. This inversion could be the result of a structural shift in the size-based PEAD, potentially due to improvements in information dissemination and liquidity among large firms.

In terms of industry segmentation, our findings show that PEAD persists in a non-uniform way, with significant long-short strategy profitability observed only in specific industries—namely HiTech, Manuf, and Shops. However, we also find that the strategy yields statistically significant negative returns in Utilities and the residual Other category, both with highly significant negative Sharpe ratios. This asymmetry implies that naïvely applying a uniform long-short strategy across all sectors may not only dilute performance but also generate persistent losses in certain market segments. These results mirror the idea that informational environments differ across sectors, and align with later research (e.g. Livnat & Mendenhall, 2006) suggesting that market reaction to earnings news varies with the clarity and interpretability of earnings signals.

These findings point to the necessity of a conditional, industry-specific investment strategy. In practice, investors would restrict the long-short implementation to industries with historically consistent, positive abnormal returns—namely, HiTech, Shops, and Manuf—while excluding or reversing positions in sectors with significant underperformance. Furthermore, investors may benefit from dynamically adjusting exposure based on updated SUE signals, sector-specific volatility, and post-2020 market conditions, which appear to have weakened the aggregate PEAD effect.

Our research highlights the limitations of evaluating size and industry in isolation. We do not explore the interaction between firm size and industry, which may partly explain the observed heterogeneity in drift strength. Future research could examine joint effects by constructing size-adjusted industry portfolios or applying multivariate regressions with interaction terms. Moreover, expanding the analysis to include further data like analyst coverage, institutional ownership, or earnings call sentiment might help to isolate the effect of earnings surprises on the market.

5. Conclusion

In this study we have attempted to empirically understand if in today's markets it is still possible to achieve a winning strategy based on standardized unexpected earnings (SUE): a failure of the market to price correctly assets after earnings announcements.

We concluded that the PEAD strategy cannot be exploited by applying it to all the assets in the three exchanges, since the average return of the strategy, with a 5% confidence level, is not significantly different from zero; moreover, the cumulative return obtained with the strategy over the 20 years of data gathered is extremely negative. This is probably due to the state of the economy after the COVID pandemic and the Ukraine war; while earlier periods appear more favorable, we lack sufficient statistical evidence to confirm consistent pre-2020 profitability. Our overall finding contradicts earlier literature, such as Bernard and Thomas (1989), suggesting a significant shift in market efficiency dynamics over time.

However, we find that the strategy remains effective in specific subsets. The application of the long-short strategy within different industries showed more robust results, allowing us to identify the sectors that produced positive abnormal returns. We noticed as well that for some industries, a reversal momentum strategy (long on Decile 1 and short on Decile 10) could be applied, since they have generated significant negative returns. At the same time, among large-cap firms, the long-short portfolio delivers positive and statistically significant returns, in contrast to the historically observed small-cap PEAD dominance. This shift may reflect structural changes in information dissemination and liquidity.

From a practical perspective, our results imply that investors seeking to exploit post-earnings announcement drift should avoid broad-market SUE-based strategies and instead focus on specific market segments. Targeting large-cap firms and select industries—such as HiTech, Manuf, and Shops—where SUE signals remain informative, may enhance performance. Conversely, applying the strategy in sectors with historically negative or insignificant outcomes could reduce or even reverse returns.

Disclaimer

This project has made use of generative AI tools, including ChatGPT and similar technologies, to support research, enhance understanding, and improve the clarity and quality of language. All content was critically reviewed and refined by the authors to ensure accuracy and academic integrity.

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